THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
KILIMANJARO REGIONAL COMMISSIONER'S OFFICE

FORM FOUR POST MOCK EXAMINATION

032/2 CHEMISTRY 2B (ACTUAL PRACTICAL)

2:30 Hours **Monday 29th September, 2025**

MARKING SCHEME

1.

(a) Table of result

wazaelimu.com

For the pipette of $25cm^3$

Titration	**Pilot**	**1**	**2**	**3**
Final reading (cm^3)	26.00	25.00	25.10	25.00
Intial reading (cm^3)	0.00	0.00	0.00	0.00
Volume used (cm^3)	26.00	25.00	25.10	25.00

06 Marks

$$\text{Average volume of acid} = \frac{V_1 + V_2 + V_3}{3}$$

$$V_a = \frac{25.00 + 25.10 + 25.00}{3}$$ **01 Mark**

$$V_a = \underline{25.03\ cm^3}$$

(b) (i) <u>$25.00cm^3$</u> of I required <u>$25.03\ cm^3$</u> of H for complete reaction

(ii) $2HCl_{(aq)} + N_2CO_{3\,(aq)} \longrightarrow 2NCl_{(aq)} + CO_{2\,(g)} + H_2O_{(l)}$ **01 Mark**

$2H^+_{(aq)} + \cancel{2Cl^-}_{(aq)} + \cancel{2N^+}_{(aq)} + CO_3^{+}{}_{(aq)} \longrightarrow \cancel{2N^+}_{(aq)} + \cancel{2Cl^-}_{(aq)} + CO_{2\,(g)} + H_2O_{(l)}$

<u>Ionic equation: $2H^+_{(aq)} + CO_3^{\pm}{}_{(aq)} \longrightarrow CO_{2\,(g)} + H_2O_{(l)}$</u> **02 Mark**

(c)Methyl orange indicator is used to determine the end point of titration by changing the colour from yellow to pink since strong acid (HCl) used hence Methyl orange works best

02 Marks

(d) Calculate the molar mass of N_2CO_3 and hence identify element N

Apply for the formula

$$\frac{M_a V_a}{M_b V_b} = \frac{n_a}{n_b}$$ **0.5 Mark**

Where:

Ma = Molarity of Acid
Va =Volume of Acid
Mb =Molarity of Base
Vb=Volume of Base
Na =Number of Acid
Nb =Number of Base **0.5 Mark**

$$2HCl_{(aq)} + N_2CO_{3\,(aq)} \longrightarrow 2NCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}$$

$N_a = 2$ $N_b = 1$

We have

Ma = 0.1M

Va =25.03 cm^3

Mb = ?

Vb=25.00 cm^3

Nb =1

Na =2

$$M_b = \frac{M_a \times V_a \times n_b}{V_b \times n_a}$$

$$M_b = \frac{0.1 \times 25.03 \times 1}{25.00 \times 2}$$

$$M_b = 0.05 \text{ M}$$

Molarity of base (Mb) = 0.05M. **03 Marks**

Then

Concentration of solution N_2CO_3 = $\frac{\text{Mass}}{\text{Volume}}$

Conc of N_2CO_3 = $\frac{2.65g}{0.5dm^3}$

Conc of N_2CO_3 = $5.3g/dm^3$ **02Mark**

But

Molar mass N_2CO_3 = $\frac{\text{Conc in } g/dm^3}{\text{Molarity } (mol/dm^3)}$

Molar mass N_2CO_3 = $\frac{5.3g/dm^3}{0.05mol/dm^3}$

- Molar mass N_2CO_3 = 106g/mol **02 Marks**

Therefore

$N_2CO_3 = 106$

$2N + 12 + (16 X 3) = 106$

$2N + 12 + 48 = 106$

$2N + 60 = 106$

$2N = 106 - 60$

$2N = 46$ **02 Marks**

$\frac{2N}{2} = \frac{46}{2}$

Atomic mass of N = 23 **02 Marks**

- Molar mass of N_2CO_3 = 106g/mol and element N is Sodium (Na) **01 Marks**

2.

S/n	Observation	Inference
(a)	White crystalline salt observed	NH_4^+, Na^+, Ca^{2+}, Zn^{2+}, Pb^{2+} May be present. Or Transition metals Fe^{2+},Fe^{3+},Cu^{2+} May be absent. **02 Marks**
(b)	Cracking sound with brown gas	NO_3^- of Pb^{2+} May be present **02 Marks**
(c)	Insoluble in cold water but soluble in hot water	Cl^- of Pb^{2+} May be present **02 Marks**
(i)	Effervescence of colourless gas evolved, which turned lime water milky	CO_3^{2-},HCO_3^- May be present **02 Marks**
(ii)	Evolution of brown fumes, which turned wet litmus paper from blue to red	NO_3^- May be present **02 Marks**
(iii)	White precipitate was formed, soluble in excess	Zn^{2+}, Pb^{2+} May be present **02 Marks**
(iv)	White precipitate was formed, insoluble in excess	Pb^{2+} May be present **02 Marks**
(v)	Brown ring was formed at the junction of the liquids	NO_3^- Confirmed **02 Marks**
(vi)	Yellow precipitate which disappeared on warming but re-appears on cooling	Pb^{2+} Confirmed **02 Marks**

Conclusion

(i) The cation of sample T is $\underline{Pb}^{2+}$ **01 Mark**

(ii) The anion of sample T is $\underline{NO_3}^-$ **01 Mark**

(iii)The chemical formula of sample T is $\underline{Pb(NO_3)}_2$ **01 Mark**

(iv) $PbNO_{3\,(aq)} + H_2SO_{4\,(aq)} \longrightarrow PbSO_{4\,(aq)} + NO_{2\,(g)} + H_2O_{(l)}$ **02 Marks**

(v) The confirmatory test for the anion present: To a small amount of sample T in a test tube, FeSO4 solution was added followed by Conc.H_2SO_4 then Brown ring was formed at the junction of the liquids hence NO_3^- Confirmed

02 Marks